



# International trade, risk and the role of banks☆☆☆



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## ABSTRACT

International trade exposes exporters and importers to substantial risks. To mitigate these risks, firms can buy special trade finance products from banks. Based on unique data with global coverage, this paper explores under which conditions and to what extent firms use these products. 15% or \$2.5 trillion of world exports are settled with letters of credit and documentary collections. Letters of credit are employed the most for exports to countries with intermediate contract enforcement, and they are used for riskier destinations than documentary collections. The 2007/2008 financial crisis affected firms' payment choices, pushing them to use more letters of credit. These patterns follow naturally from a model of payment contracts in international trade.

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## 1. Introduction

International trade is a risky activity. To manage risks and limit exposures when selling and buying abroad, firms can employ letters of credit (LC) and documentary collections (DC) offered by

banks.<sup>2</sup> These instruments became the focus of attention during the 2007/2008 financial crisis, as firms complained about a lack of supply amid global banking troubles and policy makers worried about potential adverse effects for trade. Yet, although several studies have shed light on the payment practices of individual countries and firms, it has been challenging to assess how relevant these instruments are for global trade due to a lack of accurate data with world-wide coverage.<sup>3</sup>

This paper exploits unique data from the Society for World-wide Interbank Financial Telecommunications (SWIFT), the provider of the single most important communications platform for banks, to establish new facts on the world-wide use of letters of credit and documentary collections. The data are the most comprehensive available, covering an estimated 90% of world-wide LC bank-to-bank flows. At the same time, they are very accurate as comparisons with

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<sup>2</sup> In a letter of credit, the importer's bank guarantees payment to an exporter upon proof that the goods were delivered to the importer. A DC involves ownership documents that are forwarded by the exporter's bank to the importer's bank; the importer receives the DC only upon payment. The two payment methods are illustrated in Figs. 1 and 2. For a more detailed discussion, see Section 2.

<sup>3</sup> For papers with information on the payment practices of individual countries and firms, see Antràs and Foley (2015), Hoefele et al. (2016), Ahn (2014), Demir and Javorcik (2014), Turkcan and Avsar (2016) and Del Prete and Federico (2014).

customs data show. They provide information on the number and value of LC and DC messages by location of the sending and receiving banks over an extended time period.

Letters of credit and documentary collections are relevant for exporting but they are less prevalent than suggested previously. By value of exports, LCs were used in 2011 for 13% of world exports (\$2.3 trillion), while DCs were employed for 1.8% (\$319 billion). This means that the often-cited 47% of bank-intermediated trade found in the IMF-BAFT survey has substantially overstated the size of trade finance.<sup>4</sup> The SWIFT data, through their greater coverage and representativeness, also shed new light on the United States. In 2011, 8.5% of U.S. exports were settled with LCs and 1.1% with DCs, implying that the average U.S. firm uses LCs more frequently and DCs less frequently than the food exporter studied by Antràs and Foley (2015).<sup>5</sup>

The paper documents a number of additional key facts. First, many country pairs do not use LCs at all. While 19,563 country pairs in the data traded with each other in 2010, only 7446 of or 38.1% of these exchanged at least one LC message. Zeros are especially frequent for trading pairs with Sub-Saharan African countries as importers and exporters, indicating that firms in these countries might not have the option to employ LCs due to a lack of supply.

Second, the 2007/2008 financial crisis had an impact on how international trade transactions were settled. Around the time of the Lehman bankruptcy in September 2008, the use of both LCs and DCs rose significantly. The increase in LCs was considerably larger than that of DCs, suggesting that exporters prefer LCs in an uncertain global economic environment. This finding contrasts with results in Malouche (2009) who concludes that the use of LCs and DCs did not increase after the Lehman collapse.

Third, the average size of trade transactions differs by type of payment contract. For U.S. exports, LC transactions are by far the largest with an average of \$669.7 thousand, followed by DCs (\$120.4 thousand). Transactions that do without bank intermediation (cash-in-advance and open account) are on average much smaller (\$37.4 thousand). This ordering is consistent with substantial fixed costs in the provision and use of trade finance products.

The paper also explores in depth the role of risk for firms' payment choices, using a country's quality of legal institutions as a proxy for the inverse risk that companies face when trading internationally. The regression analysis delivers four results. The relationship between letters of credit and destination country risk is hump-shaped. In other words, LCs are used the most for exports to countries with intermediate contract enforcement.<sup>6</sup> The use of DCs also has a hump-shaped relationship with destination country risk. Compared to DCs, LCs are used for riskier destinations. Finally, the use of LCs and DCs also first rises and then falls with contract enforcement in the source country.

The documented facts can be rationalized by a theoretical model of payment choice based on Schmidt-Eisenlohr (2013), detailed in the second part of the paper. Specifically, we extend the framework in the aforementioned paper by including documentary collections in firms' contract choice set and by refining the formulation of the letter of credit fee. While several papers have modified the model in Schmidt-Eisenlohr (2013), the focus of these works has been on

aspects such as heterogeneity across firms (Glady and Potin 2011) and products (Hoefele et al. 2016), the role of industry-level competition (Demir and Javorcik 2014), or dynamics (Antràs and Foley 2015). Only the modifications in this paper allow the model to match all of the empirical results presented here.<sup>7</sup>

As mentioned, LCs tend not to be used for either the least risky or most risky destinations. The basic intuition for this result, made explicit in the model, is that the value of risk mitigation through bank intermediation is offset to a degree by the cost of the intermediation. Because banks can reduce but cannot eliminate the risk of a trade transaction, the fees they charge rise with the remaining risk they take on. For the riskiest destination countries, bank fees are so high that exporters prefer cash-in-advance. In that case, the importer pays before the exporter produces, and payment risk is eliminated. Similarly, LCs are not used for low-risk destinations; for those transactions, the exporter can save on bank fees by bearing the risk itself.

A calibration exercise shows that the model can match the facts not only qualitatively but also quantitatively. In particular, the model can fit the aggregate use of LCs and DCs in U.S. exports as well as the shares of these instruments across destination countries with different levels of contract enforcement. An increase in the riskiness of destination countries generates the shift in the use of letters of credit observed during the 2007/2008 financial crisis.

Besides contributing to the growing literature on payment forms in international trade, the paper relates to the literature on financial conditions and comparative advantage.<sup>8</sup> It is also closely linked to a set of papers that explore how shocks to the supply of trade finance affect exports (Amiti and Weinstein 2011, Paravisini et al. 2015, Del Prete and Federico 2014, Ahn 2013, and Niepmann and Schmidt-Eisenlohr 2013).<sup>9</sup>

The remainder of the paper is structured as follows. Section 2 provides background information on payment terms in international trade. Section 3 introduces the SWIFT data. Section 4 documents key facts. Section 5 presents the regression analysis with a focus on the role of risk. Section 6 develops a model of payment choice that explains the empirical patterns and reports results from calibration and simulation exercises. Section 7 discusses implications of our analysis.

## 2. A brief primer on trade finance

Trade is risky and takes time. For this reason, it matters how the payment of a transaction is organized. The four most important payment forms in international trade are open account, cash-in-advance, LCs and DCs. Each type differs in the allocation of risk, the financial cost and the extent of bank involvement. Table 1 summarizes key differences between the four choices.

### 2.1. Cash-in-advance

Under cash-in-advance (CIA), the importer pays first for the good and the exporter produces the good after receiving payment. Since sales revenues only realize later, the importer has to pre-finance the

<sup>4</sup> Part of the discrepancy may be due to a wider interpretation of bank-intermediated trade by survey respondents. See IMF-BAFT (2011) and Asmundson et al. (2011).

<sup>5</sup> Antràs and Foley (2015) show that the firm's use of LCs decreases with the degree of contract enforcement in the destination country and that the firm is more likely to use an LC when interacting with a new customer.

<sup>6</sup> Glady and Potin (2011), who use SWIFT message data for the year 2006, find that the use of LCs increases in country risk as predicted by their model of payment contract choice. A close inspection of their regression results also indicates a hump-shaped relationship, an aspect that is, however, not discussed by the authors.

<sup>7</sup> Ahn (2014) develops an alternative model that explains the popularity of open account transactions by their self-liquidating and recourse nature. In Ahn (2011), an earlier version of the aforementioned paper, the focus is on the effect of changes in aggregate default risk on the ratio of exports to domestic sales; in that model, LCs become relatively less attractive in a crisis because risks increase both for importers and banks.

<sup>8</sup> See, in particular, Beck (2003) and Manova (2013) on how financial development affects trade patterns.

<sup>9</sup> Niepmann and Schmidt-Eisenlohr (2013) also employs the SWIFT data analyzed here. Instead of studying payment choices across countries, it uses the SWIFT data to show that quarterly trade finance exposures from U.S. FFIEC 009 reports strongly correlate with letter of credit flows.

**Table 1**  
Tradeoffs between different payment forms.

|          | OA   |                 | CIA  |                 | DC   |                 | LC   |                 |
|----------|------|-----------------|------|-----------------|------|-----------------|------|-----------------|
|          | Risk | Financial costs | Risk | Financial costs | Risk | Financial costs | Risk | Financial costs |
| Exporter | ++   | +               | –    | –               | +    | ++              | –    | +               |
| Importer | –    | –               | ++   | +               | –    | –               | –    | +               |

Note: The table illustrates how the four payment contracts (open account, cash-in-advance, letters of credit, documentary collections) differ in terms of their implied allocations of risk and financial costs. The notation has the following interpretation. Risk: ++ all risk, + some risk, – no risk. Financial costs: ++ interest expenses and fees, + interest expenses or fees, – zero costs.

payment to the exporter. At the same time, the importer faces the risk that the exporter does not deliver or supplies products of low quality. The importer, thus, bears both the financial cost and the risk of the transaction.

## 2.2. Open account

With an open account (OA), the exporter delivers the good and the importer pays upon receipt. Now the exporter bears the financial cost and the risk of the transaction. On the one hand, the exporter has to pre-finance working capital since it produces before being paid. On the other hand, the importer may delay payment or may not pay at all. Exporters can cover all or some of this risk through trade credit insurance. In addition, they may sell some or all of their claims to factoring companies to raise working capital.

Banks can be involved in both cash-in-advance and open account transactions. They typically help with financing the firms' pre-payment or working capital and may assist exporters in finding insurance providers or factors.

## 2.3. Letter of credit

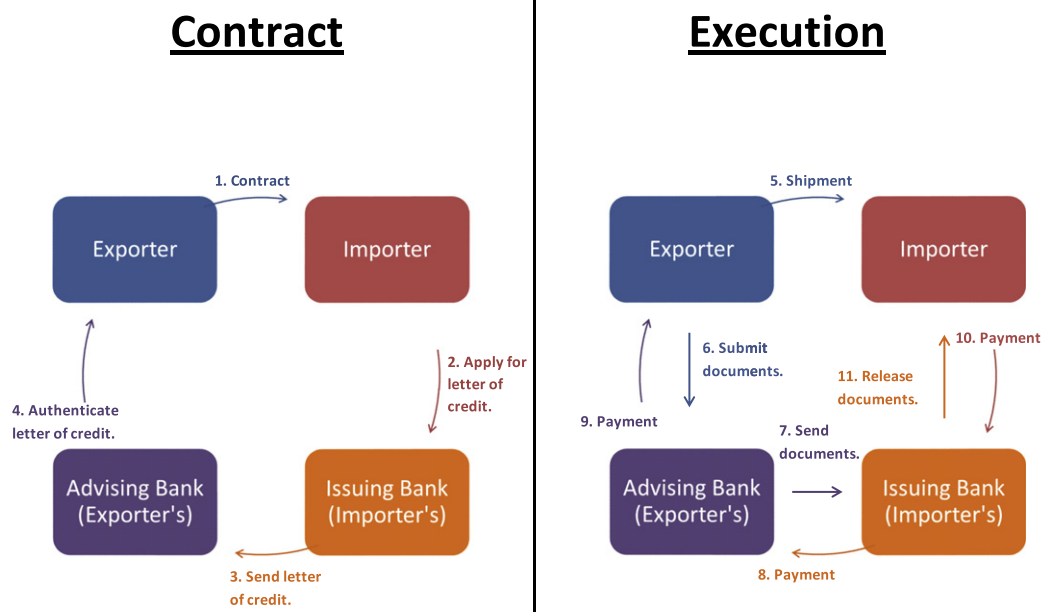
LCs are the most common trade finance product provided by banks. The importer initiates the LC transaction (Fig. 1) by having its bank issue the instrument to the exporter. The LC guarantees that the issuing bank will pay the agreed contract amount when the

exporter proves that it delivered the goods, for example, by providing shipping documents confirming the arrival of the goods in the destination country. To cover the risk that the issuing bank will not pay, an exporter may have a bank in its own country confirm the LC, in which case the confirming bank agrees to pay the exporter if the issuing bank defaults.

## 2.4. Documentary collection

Another way for banks to assist firms in international trade, although less common, is DCs. In contrast to an LC, a DC does not involve payment guarantees. Instead, the exporter's bank forwards shipping/ownership documents to the importer's bank; the documents, which transfer the legal ownership of the traded goods to the importer, are handed to the importer only upon payment for the goods (Fig. 2).

A DC provides less security to the exporter than an LC. In the latter case, an exporter is paid by the issuing bank/confirming bank upon proof of delivery regardless of whether the importer paid. With a DC, the exporter is remunerated only if the bank receives the payment from the importer. As mentioned, trade credit insurance is another way for exporters to handle risk in international trade. An important difference is that credit insurance shifts the risk from the exporter to the insurer, whereas LCs and DCs reduce the risk inherent in the transaction since they increase the importer's incentives to pay. LCs and DCs are short-term trade finance instruments. According to the International Chamber of Commerce Trade Register, the average



**Fig. 1.** How a letter of credit works.

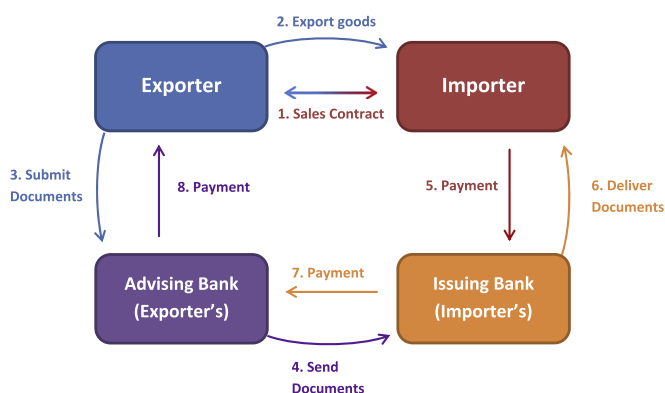


Fig. 2. How a documentary collection works.

maturity of a confirmed LC is 70 days, while the average maturity of an importer LC is about 80 days (see ICC 2013).

### 3. A new data set from SWIFT

#### 3.1. Data sources in previous studies

Data on trade finance are scarce, especially those with world-wide coverage, and are presented only in a few papers. International organizations were first to collect information through surveys. In the IMF-BAFT survey, for example, bankers were asked to assess how world trade was financed (IMF-BAFT 2011). In 2014, the Committee on the Global Financial System under the umbrella of the Bank for International Settlements (BIS) gathered various trade finance data collected by BIS member countries to provide cross-country estimates (see BIS 2014). Antràs and Foley (2015) explore data from one large U.S. food exporter who disclosed how it settles its international shipments. More recently, comprehensive customs data for individual countries has become available. Ahn (2014) documents the use of trade finance in Colombian and Chilean imports, while Demir et al. (2014) study Turkish exports.

Since it is hard to know how representative and accurate surveys are, it remains an open question how much of trade is intermediated by banks worldwide, with figures differing substantially across sources. There is also only limited knowledge of cross-country differences in the use of trade finance instruments and of the factors that drive them. This paper contributes to filling these gaps by exploiting a new data set derived from SWIFT messages sent by banks around the world.

#### 3.2. The SWIFT data

The Society for Worldwide Interbank Financial Telecommunications (SWIFT) network provides a communications platform through which financial institutions exchange standardized financial messages. SWIFT reports that more than 10,500 corporations and financial institutions in 215 countries use this platform and estimates that about 90% of worldwide LC bank-to-bank flows go through the SWIFT network. While there is no data that allows us to directly verify this global number, we show below that for those countries where detailed data is available, SWIFT messages basically capture all trade paid for with LCs. In addition, there is ample anecdotal evidence

that there is no substitute for the SWIFT system.<sup>10</sup> According to the SWIFT Institute, even messages between different parts of multinational banks are often on SWIFT.<sup>11</sup> This allows banks to use SWIFT's traffic analysis tools and the marginal costs of sending messages are relatively low.

When a bank in the importer's country issues an LC, it sends a so-called MT700 message to the advising bank in the exporter's country; when it releases a payment related to a DC, it sends an MT400 message. The message specifies the terms of the LC or the payment, including the names of the trading parties, the banks involved and the goods traded. It can happen that, at some point, a bank wants to cancel or amend the terms of an LC or a DC. For this purpose, it can send message types MT707 and MT407. According to the SWIFT Institute, this is a relatively rare event. Banks also issue so-called standby LCs using message type MT760. Standby LCs provide a general backstop against non-performing counterparties and work more like guarantees or insurance. They seem to be used more for domestic transactions and services as well as long-term international investments than for standard international trade. Unfortunately, SWIFT does not disclose information on any of these additional message types.

Due to confidentiality, the SWIFT Institute provides the messages data aggregated up to the country-pair level. Our data set contains the number of MT700 and MT400 by sender and recipient country at a monthly frequency from 2003 to 2012. From the fourth quarter of 2010 onward, we also know the total value of the messages in U.S. dollars. Because the value data is available for only a relatively short period, and message counts and amounts are highly correlated, we mainly use the count data for the analysis.

Unfortunately, we do not observe messages by industry. Since there is evidence that the use of trade finance instruments differs across industries (see, for example, Hoefele et al. 2016 and Demir and Javorcik 2014), some of the differences in their use across countries could be driven by the industry composition of trade. Due to the nature of the data, we cannot investigate this issue and have to accept this limitation.

#### 3.3. Accuracy of the SWIFT data

The reader might wonder how accurate the SWIFT data are. Fortunately, we have a way to check this. The three data sets derived from customs records presented in Ahn (2014) and Demir et al. (2014) deliver arguably the most reliable and representative numbers since exporters and importers have to declare at the transaction level which payment form they employ. For comparison, we therefore calculate the same moments as reported in the aforementioned studies for Chile, Colombia and Turkey based on the SWIFT data for the year 2011. Results are shown in Table 2. For all three countries, the SWIFT data deliver numbers almost identical to the customs data. So the SWIFT data seem to provide an accurate measure of aggregate trade finance activity.

Since we are also interested in the country-pair dimension, we go one step further and compare numbers also at the bilateral level. To this end, we regress SWIFT LC shares on the LC shares from customs data. We do not allow for a constant in these regressions. Columns (1), (3) and (5) of Table 3 report results for all country pairs for which

<sup>10</sup> For example in the context of recent discussions by Russia about setting up a BRICS alternative to SWIFT, the Deputy Chairman of the Central Bank of the Russian Federation stated in mid-2015: 'Seriously speaking, there is no (alternative) to SWIFT in the world right now.' See Forbes (2015) for details.

<sup>11</sup> The SWIFT Institute is an entity set up by SWIFT that, among others, funds research and provides access to data derived from the SWIFT messaging system.



**Table 2**  
Data on LC and DC use.

| Country                          | Ex/Im | LC                       | DC          | Source                           |
|----------------------------------|-------|--------------------------|-------------|----------------------------------|
| Survey data/individual firm data |       |                          |             |                                  |
| World                            | —     | 47 <sup>1)</sup><br>13.0 | —<br>1.8    | IMF-BAFT (2011)<br>SWIFT         |
| US <sup>2)</sup>                 | Ex    | 5.5<br>8.5               | 10.7<br>1.1 | Antràs and Foley (2015)<br>SWIFT |
| Customs data                     |       |                          |             |                                  |
| Colombia                         | Im    | 5<br>5.0                 | —<br>0.01   | Ahn (2014)<br>SWIFT              |
| Chile                            | Im    | 10<br>10.1               | —<br>0.3    | Ahn (2014)<br>SWIFT              |
| Turkey                           | Ex    | 15<br>15.7               | —<br>2.7    | Demir et al. (2014)<br>SWIFT     |

Note: The table summarizes results from previous studies and surveys on the shares of LCs and DCs in international trade and compares them to values calculated from SWIFT data. All numbers indicate the percentage share in the value of total trade (either from customs data, individual firm data or from guesses of survey participants). The Ex/Im column specifies if a data point refers to exports or imports. All numbers are for 2011 except those from Antràs and Foley (2015), who report averages from 1996 to 2009. 1) The IMF-BAFT survey does not ask about LCs or DCs but asks about bank-intermediated trade. Respondents may subsume additional payment forms under this category (IMF-BAFT 2011). 2) Antràs and Foley 2015 use information from one U.S. food exporter while SWIFT covers total U.S. exports.

information is available both in the customs data and in the SWIFT data.<sup>12</sup>

There are several reasons why the estimated coefficients are not equal to one. First, the customs data likely suffer from some form of measurement error.<sup>13</sup> Columns (2) and (4) support this. In these columns, regressions are run for the top 15 source countries of Chile and Colombia, respectively. The fit improves when only large suppliers are included and the coefficients get closer to 1. For Turkey (column (6)), the coefficient gets slightly larger, while the fit worsens marginally. Another reason for a discrepancy is that sometimes banks involved in the transaction are located in a third country, i.e. in a country different from the source or destination country of the shipped goods. In this case, SWIFT message flows at the bilateral level might not perfectly match up with the respective bilateral trade flows. These third countries appear to be primarily tax havens and offshore financial centers. The LC flows observed from and to these countries are large relative to the trade flows. To account for this issue, we exclude offshore financial centers and tax havens from the regression analysis that uses bilateral message flows. All results are robust to the inclusion of these countries. We do include flows to and from tax havens and offshore centers, however, when calculating the aggregate shares at the country level reported in Table 2.<sup>14</sup>

A third reason for a coefficient other than 1 is that trade occurs in different currencies and that the method to convert trade flows into U.S. dollar values may differ across data sets. To summarize, while the match is not perfect, the results show that SWIFT data is highly informative about the use of trade finance instruments not only in the aggregate but also at the bilateral level.

A closer look at the figures for the United States uncovers some key differences to Antràs and Foley (2015). LCs are used more frequently and DCs less frequently than what the data from the food producer studied in the aforementioned paper would suggest. The firm relies on LCs for 5.5% and on DCs for 11% of its shipments. In contrast, the SWIFT data show that 8.5% of U.S. exports are settled with LCs but only 1.1% with DCs.<sup>15</sup> Our data set also provides evidence on the use of LCs and DCs in U.S. imports, information that was not available previously. The share of LCs in U.S. imports is 1.9%. The share of DCs is 5.2%.

## 4. The facts

### 4.1. The prevalence of LCs and DCs in world trade

With the external validity of the SWIFT data established, this section provides new facts on the use of payment forms in international trade. In particular, we provide the first reliable estimates on the prevalence of LCs and DCs in world trade. The statistics reported below are based on observed trade values as well as LC and DC message values in 2011 if not mentioned otherwise. The underlying trade data are from the IMF Directions of Trade Statistics.

#### 4.1.1. LCs and DCs in world trade

According to the SWIFT data, LCs cover about 13% or \$2.27 trillion of world trade. DCs are employed less frequently, for only 1.8% or \$319 billion of world exports and imports. These numbers are far off those resulting from the IMF-BAFT survey, which suggested that 47% of world trade were bank-intermediated (IMF-BAFT 2011). Part of this discrepancy may be driven by the fuzzy definition of trade finance in the survey. Survey respondents might have included other trade finance products but LCs and DCs in their estimates. In any case, this often-cited number has led to an overstatement of the importance of LCs and DCs in world trade.

Fig. 3 shows the use of LCs and DCs by the top 10 world importers and exporters in 2011, revealing substantial heterogeneity across countries. For example, 36.4% of Chinese imports are based on LCs while only 4.9% of Japanese imports rely on this instrument; DCs are employed for 4.7% of Korean but only for 0.004% of Russian exports. Heterogeneity is even larger at the country-pair level. In 2011, the standard deviations of country-pair-level LC and DC shares were 0.29 and 0.33, respectively.<sup>16</sup>

#### 4.1.2. LCs and DCs by income regions

Many institutions provide trade finance support to low income countries, based on the view that trade finance is particularly useful for these countries but often difficult to obtain. To shed some light on this issue, we consider the use of LCs and DCs by region.

We split countries into five income regions following the World Bank classification and create a separate group for offshore financial centers and tax havens. Fig. 4 displays the use of LCs and DCs in imports on the left and exports on the right by these six groups of countries. Not considering tax havens and offshore financial centers, LCs are most prevalent in imports to low-income countries (27.3%), while they are used only rarely for imports to high-income

<sup>12</sup> We exclude Iran in the regression for Turkey because sanctions forced SWIFT to disconnect Iran from its network in April 2012.

<sup>13</sup> For example, in the Colombian data, firms choose between 11 different payment types. Ahn (2014) drops 4 of them when computing his statistics.

<sup>14</sup> A list of tax havens and offshore financial centers can be found in the data appendix. 24.7% (36.8%) of LC flows in terms of value have an offshore center as importer (exporter). 55.6% of LC flows have an offshore center either as exporter or importer. If Switzerland, Hongkong and Singapore are not designated as offshore centers, these shares fall to 5.4%, 7.6% and 12.9%, respectively. Results based on a sample with tax havens and offshore financial centers are reported in Tables B.12 and B.13 in the online appendix.

<sup>15</sup> One reason for these differences could be heterogeneity in LC and DC use across industries. For example, a more regulated industry like the food industry may rely more on DCs because these simplify the transfer of documentation, for example, regarding certification or source of origin.

<sup>16</sup> Standard deviations are calculated using observations with positive LC and DC flows only, with shares winsorized at 100%. In the online appendix, we provide a full list of countries and their LC and DC intensities in 2013. The data underlying these tables is also available for download at [https://sites.google.com/site/timschmidteisenlohr/tf\\_data](https://sites.google.com/site/timschmidteisenlohr/tf_data).

**Table 3**  
SWIFT data and Customs data.

|                  | Chile                |                      | Colombia            |                     | Turkey               |                     |
|------------------|----------------------|----------------------|---------------------|---------------------|----------------------|---------------------|
|                  | (1)                  | (2)                  | (3)                 | (4)                 | (5)                  | (6)                 |
| LC share Customs | 0.600***<br>(0.0601) | 0.755***<br>(0.0902) | 0.538***<br>(0.111) | 0.905***<br>(0.184) | 0.715***<br>(0.0536) | 0.742***<br>(0.123) |
| Observations     | 43                   | 15                   | 26                  | 15                  | 69                   | 15                  |
| R-squared        | 0.703                | 0.833                | 0.486               | 0.634               | 0.724                | 0.723               |

Note: The table shows how the SWIFT data compares with customs data from Columbia, Chile and Turkey. First, we compute the share of import transactions that use LCs by source country for Chile, Columbia and Turkey based on the SWIFT data. Then, we obtain the equivalent shares from country studies that use customs data. Finally, we regress the SWIFT shares on the shares obtained from customs data for each importing country separately. Columns (1), (3) and (5) present the results based on the full matched samples. Columns (2), (4) and (6) report results on imports from the 15 most important source (or destination) countries in each matched sample. Samples for columns (5) and (6) exclude Iran. Standard errors are in parentheses. \*, \*\* and \*\*\* denote significance at the 10%, 5% and 1% level.

countries (4.9%). DCs are hardly ever employed in imports to low-income countries (0.1%), while they are most common in exports by lower-middle income countries (around 2.9%). As mentioned earlier, LC and DC traffic in offshore financial centers and tax havens appear disproportionate relative to these countries' total exports and imports, probably stemming from the fact that banks in these countries provide trade finance services for parties outside of the country.

As work by Helpman et al. (2008) shows, there is a large number of country pairs in the world that exhibit zero trade flows. Similarly, there is a large number of country pairs between which no LC and DC messages are sent. In 2010, there were 19563 country pairs with positive trade flows, of which 7446(5191) pairs exchanged LC (DC) messages.<sup>17</sup> This means that only 38.1 (26.5) percent of country-pairs with positive trade flows used LCs (DCs) in that year. At the same time, however, country pairs with positive LC (DC) message flows accounted for 96.7 (90.7) percent of total trade in the data.

Fig. 5 illustrates that there is substantial variation in the number of unidirectional country pairs that trade with each other but have zero LC traffic. For example, the first bar in the chart displays the fraction of country pairs with South Asian countries as importers for which SWIFT recorded at least one LC message in 2010. While roughly 55% of country pairs with South Asian countries as importers or exporters send or receive LC messages, more than 70% of country pairs with Sub-Saharan African countries as importers or exporters have zero LC traffic.

This finding supports the view of international institutions like the World Bank and the Asian Development Bank that for many country pairs trade finance products are not available, probably due to the presence of substantial fixed costs of offering trade finance services for banks. The provision of LCs and DCs requires a well-functioning banking network between the source and the destination country, which takes time and is costly to establish. To the extent that access to trade finance is beneficial for countries, international development banks might have a role in supporting the formation of additional bank links.

#### 4.2. Changes in the use of LCs and DCs over time

Fig. 6 shows the use of LCs and DCs in world trade over time.<sup>18</sup> Chart (a) on the left plots the number of LC and DC messages relative to real world exports from 2003 to 2012. There is a secular decline in the use of both trade finance instruments over this

period. The vertical dotted lines in the charts indicate 2008q3, the quarter when Lehman collapsed and the financial crisis deepened. Both LC and DC messages saw a strong pick-up at that time. The increase in LCs was much larger compared to DCs. Chart (b) on the right illustrates this clearly. It plots the moving average of the share of LC messages in total trade finance messages. Until 2008q3, the fraction of LCs was relatively stable at around 52%. From 2008q3 onward, it moved quickly to reach a share of 61%. While some of the rise might be explained by the policy initiatives of the G20, the pattern still suggests that firms shifted towards LCs to limit their risk exposures during the crisis. The finding contrasts with survey evidence reported in Malouche (2009) who concludes that DC and LC shares did not increase in the wake of the Lehman collapse.<sup>19</sup> It is consistent with the notion that there was an insufficient supply of LCs during this time since demand was likely elevated and, according to surveys, the cost of trade finance rose sharply.

#### 4.3. Average transaction sizes

Importers and exporters have to pay a fee for LCs and DCs. Because a part of the fee is fixed, covering document handling, screening and monitoring costs, the attractiveness of these payment forms should increase with the size of an export transaction. While we do not observe the transaction value of single LCs and DCs, we can compute the average size of LC and DC transactions based on the total value of LCs and DCs and the corresponding number of SWIFT messages by importing and exporting country. To compare the average size of these transactions to the average trade transaction, additional information is needed. While not easily available for all countries, we obtain data on the number of shipments (so-called cards) for U.S. trade from the United States Census Bureau.

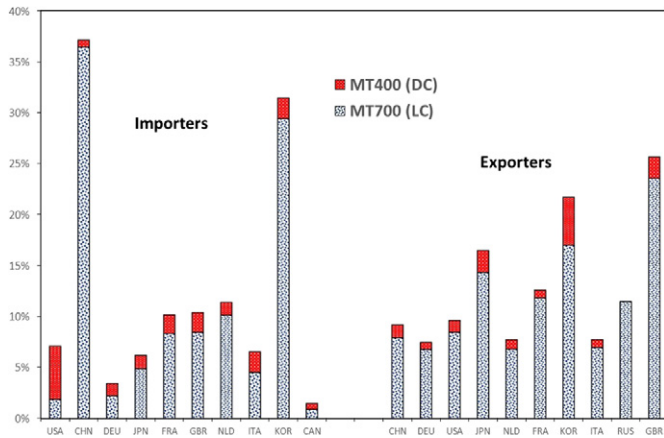
In 2012, the average value of a U.S. export transaction that employed an LC was \$669.7 thousand.<sup>20</sup> The average value of those using DCs was \$120.4 thousand. The average value of a U.S. export transaction –regardless of the underlying payment form– was \$40.3 thousand. Thus, an average LC transaction is more than sixteen times as large as an average trade transaction, while an average DC transaction is roughly three times as large. The ranking survives even when

<sup>19</sup> Malouche (2009) is based on evidence from a 2009 World Bank survey, in which the majority of 264 developing country firms responded that they were not changing their method for trade transactions.

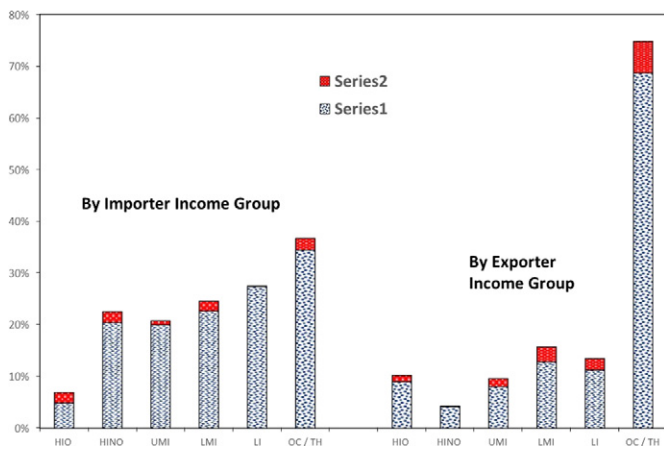
<sup>20</sup> To compare transaction sizes across payment types, we restrict the sample to countries for which each payment type is observed and exclude tax havens and offshore financial centers. This reduces the number of countries to 60. Results are very similar when we calculate the numbers for the unbalanced data set and include offshore financial centers and tax havens.

<sup>17</sup> Country pairs are defined unidirectionally, so exports from the United States to China and exports from China to the United States are counted separately.

<sup>18</sup> We thank an anonymous referee for the suggestion to explore changes in the use of LCs during the financial crisis.



**Fig. 3.** The use of LCs and DCs by the top 10 world exporters and importers. Note: The chart has the top 10 world importers on the left and the top 10 world exporters on the right. For each of these countries, it displays the share of international shipments that is settled with LCs or DCs in 2011, based on SWIFT message values and export values in that year.



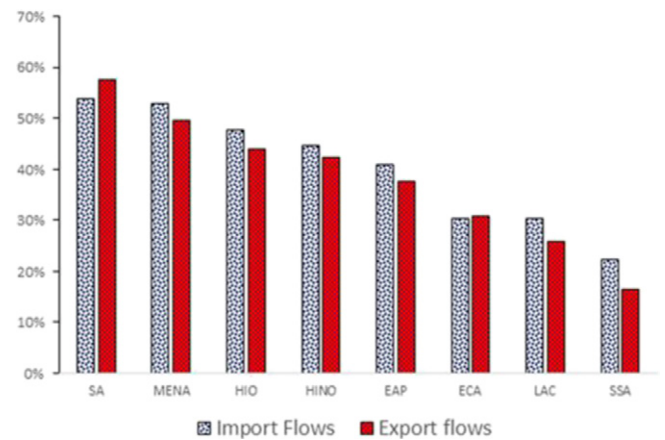
**Fig. 4.** The use of LCs and DCs by income region. Note: The chart shows the share of international trade transactions that use LCs and DCs by importer and exporter region in 2011. Countries are grouped according to their income and status as offshore financial center or tax haven. The six groups are: HIO – high-income OECD countries; HINO – high-income non-OECD countries; UMI – upper middle income countries; LMI – lower middle income countries; LI – lower income countries; OC/TH: tax havens or offshore financial centers.

average transaction sizes are analyzed for each importing country separately.<sup>21</sup>

## 5. Regression analysis

As discussed earlier, LCs and DCs reduce the risk of transactions for exporters and importers. In this section, we explore the role of both exporting and importing country risk for the use of LCs and DCs formally, based on regression analysis. The risk that firms face when trading internationally decreases with the quality of legal institutions in the trading partner's country so we proxy the inverse of country risk by contract enforcement.

<sup>21</sup> For 95% of U.S. export destinations, LCs have a higher average value than DCs. For 100 (80) percent of the countries, LCs (DCs) have a higher average value than transactions that use neither LCs nor DCs. Figure C.1 in online appendix C displays kernel density estimates of the average transaction size by country in 2012. It illustrates the higher transaction sizes of LCs and DCs relative to other payment forms.



**Fig. 5.** Positive SWIFT flows by region. Note: The chart shows the fraction of unidirectional country pairs for which we observe positive LC flows by geographic region in 2010 conditional on positive trade flows. For example, the first (second) bar shows the share of country pairs with South Asian countries as importers (exporters) that had positive LC traffic in that year. The acronyms stand for: SA – South Asia; MENA – Middle East and North Africa; HINO – high-income non-OECD; high-income OECD; EAP – East Asia and Pacific; ECA – Eastern Europe and Central Asia; LAC – Latin America and Caribbean; SSA – Sub-Saharan Africa.

The analysis delivers two main results. First, the use of LCs and DCs is hump-shaped in contract enforcement both in the importing and the exporting country. Second, LCs are used for riskier destinations than DCs. These patterns can be explained by payment choice theory as we show in Section 6.

### 5.1. The role of importer and exporter risk

#### 5.1.1. Importer risk

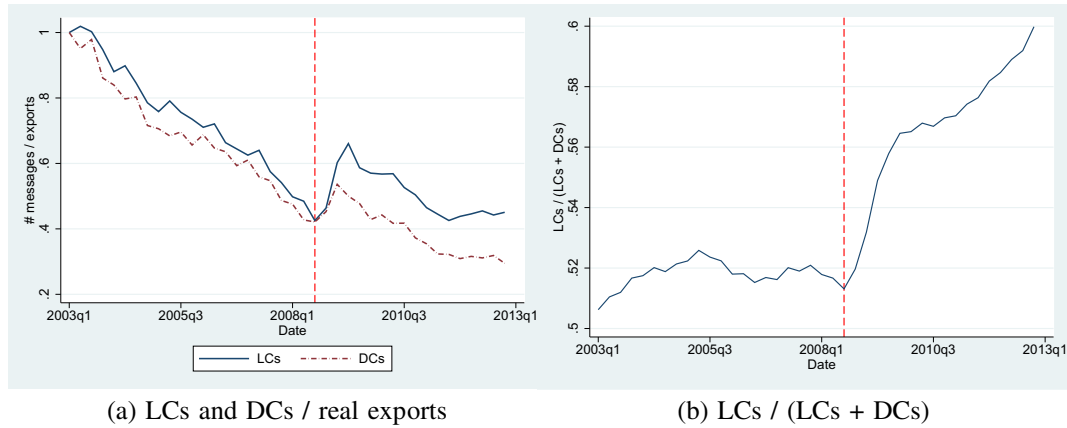
We start by studying the link between the use of LCs and DCs and importing country risk, estimating the following equation:

$$\begin{aligned} \log(Y_{ijt}) = & \beta_1 \log(\exp_{ijt}) + \beta_2 \log(\text{distance}_{ij}) + \beta_3 \text{law}_{it} + \beta_4 (\text{law}_{it})^2 \\ & + \beta_5 \log(\text{GDP per capita}_{it}) + \beta_6 (\log(\text{GDP per capita}_{it}))^2 \\ & + \beta_7 \log(\text{fin. developm}_{it}) + \beta_8 \text{LC requirement dummy}_{it} \\ & + \beta_9 \text{bilateral controls}_{ij} + \alpha_{jt} + \epsilon_{ijt}. \end{aligned} \quad (1)$$

$Y_{ijt}$  stands for the number either of LC or of DC messages sent to banks from importing (destination) country  $i$  to exporting (source) country  $j$  in year  $t$ . The dependent variable is regressed on the log of exports, on the log of distance, and on contract enforcement in country  $i$ , the destination of the shipped goods. We use as a proxy for contract enforcement the rule of law index from the World Bank's World Governance Indicators ( $\text{law}_{it}$ ). Since the index takes negative values for some countries, it is mapped on the interval between 0 and 1, where 1 corresponds to the highest degree of the rule of law observed. The regression also includes rule of law squared to allow for a non-linear relationship.<sup>22</sup> It further controls for the destination country's stage of financial development. In line with the extensive literature on the topic, we use as a measure of financial development the variable *private credit by financial institutions over GDP* from the World Bank's Financial Structure Database (see Beck 2003 and Manova 2013).

Because some countries require firms to use LCs when trading internationally, the equation includes a dummy variable that takes the value of 1 when such a requirement is in place in country  $i$  at

<sup>22</sup> In online appendix F, we show that semi-parametric estimation that allows for an entirely flexible form delivers very similar results.



**Fig. 6.** The use of LCs and DCs over time. Note: The figure shows the use of LCs and DCs over time. Panel (a) depicts the ratios of LCs and DCs over real exports (nominal exports deflated with the U.S. GDP deflator). The ratios are indexed to 1 in 2003q1. Panel (b) shows the ratio of LC messages over the sum of LC and DC messages. To filter out short-term fluctuations, we smooth the ratio in panel (b), showing a 5-quarter moving average with weights 1/2/3/2/1. The vertical line in both panels indicates 2008q3.

time  $t$ .<sup>23</sup> Log of GDP per capita and log of GDP per capita squared are also added as regressors, to ensure that the effect of contract enforcement is not due to omitted factors correlated with a country's overall development. Moreover, all regressions control for a set of country-pair specific variables (contiguity, common colonizer, common language, time difference) and exporter-time fixed effects. Standard errors are clustered by importing country. Note that we do not include destination country fixed effects, so identification comes primarily from cross-country variation in contract enforcement. This is the correct specification for testing the aggregate non-linear relationship that we are interested in.<sup>24</sup>

Table 4 presents evidence on the relationship between SWIFT traffic and the degree of contract enforcement in the importing country. Column (1) shows the results for LCs. Column (2) is for DCs. In both columns, the coefficient of the linear rule of law term is positive and significant, while the coefficient of the squared term is negative and significant. Thus, the relationship between a destination's rule of law and the use of the respective trade finance instrument in world trade is non-linear. The coefficients of financial development and distance are also highly significant and positive, indicating that LCs and DCs are used more for exports to more financially developed countries as well as for long-distance trade.

To explore differences in the relationship between the use of DCs and LCs and a country's rule of law, we estimate the following pooled regression:

$$\begin{aligned} \log(Y_{mijt}) = & \beta_1 \log(\exp_{ijt}) + \beta_2 \log(\text{distance}_{ij}) + \beta_3 \text{law}_{it} + \beta_4 (\text{law}_{it})^2 \\ & + \beta_5 \text{DC message dummy}_{mijt} + \beta_6 \text{law}_{it} \\ & \times \text{DC message dummy}_{mijt} + \beta_7 \log(X_{i(j)t}) + \alpha_{jt} + \epsilon_{ijt}. \end{aligned} \quad (2)$$

Now  $\log(Y_{mijt})$  stands for the number of messages of type  $m$  sent from importing country  $i$  to exporting country  $j$  in year  $t$ . The regression includes a dummy that takes the value of 1 if the message refers to a

DC and zero if the message is related to an LC, as well as interactions between the dummy and the rule of law variables.  $X_{i(j)t}$  collects the control variables specified earlier.

Column (3) of Table 4 shows that the coefficients of the linear term differ significantly for DCs and LCs, being larger for DCs. Column (4) delivers no evidence for systematic differences in the quadratic term coefficients between message types. Panel (a) of Fig. 7 plots the relationship between the use of LCs and DCs, respectively, and the rule of law index based on coefficients in column (3). LCs are employed the most for exports to countries with intermediate contract enforcement. Their use peaks at a rule of law index of 0.52, equivalent to Panama's in 2012. The use of DCs also first rises and then fall with the importer's rule of law. The share of exporters that use DCs is highest for a rule of law index of 0.86, which corresponds to Chile's in 2012. Thus, the use of DCs in exports reaches its maximum at higher contract enforcement than the use of LCs.

To see that rule of law is an economically important determinant of firms' payment choices, consider the following example based on the estimated coefficients in column (3). If Guatemala (normalized rule of law of 0.34) had the same rule of law as Brazil (normalized rule of law of 0.55 in 2012), the number of LC messages sent from Guatemala would increase by 19.8 percent. In contrast, if Brazil had the same rule of law as Israel (normalized rule of law of 0.77 in 2012), LC messages sent from Brazil would decline by 36.3 percent. The moves from Guatemala to Brazil and from Brazil to Israel correspond to roughly one standard deviation of the rule of law index.

### 5.1.2. Exporter risk

We conduct a parallel regression analysis to study the role of contract enforcement in the exporting country. Regressions now include exporter variables and importer-year-fixed effects. Standard errors are clustered by exporting country. Table 5 shows the results, which resemble those for destination country risk. SWIFT traffic also displays a non-linear relationship with source country rule of law. Moreover, greater financial development of the exporting country promotes the use of trade finance instruments. The regressions provide mixed evidence on whether source country rule of law has a differential effect on the use of LCs versus DCs. The interaction term between the message type dummy and rule of law in column (3) is insignificant. It becomes marginally significant and negative in column (4) where an interaction term with rule of law squared is added. Panel (b) of Fig. 7 plots the estimated relationship between LC/DC traffic and rule of law in the exporting country based on column (3).

<sup>23</sup> Information on countries' documentation requirements is from the IMF's AREAAR Database.

<sup>24</sup> Including destination country fixed effects would instead deliver an estimate of a local non-linear effect, that is, the model would be misspecified for estimating the relationship we want to study. As McIntosh and Schlenker (2006) put it: 'The marginal effects of a relationship which is globally quadratic in  $X$  must depend on the non-demeaned value of  $X$ , and so inherently cannot be identified in deviations from a group mean.' See their paper for technical details.



**Table 4**

LCs, DCs and destination country rule of law: intensive margin.

|                                                                        | (1)                         | (2)                         | (3)                                | (4)                                |
|------------------------------------------------------------------------|-----------------------------|-----------------------------|------------------------------------|------------------------------------|
|                                                                        | $\log(\# \text{ LC}_{ijt})$ | $\log(\# \text{ DC}_{ijt})$ | $\log(\# \text{ messages}_{mijt})$ | $\log(\# \text{ messages}_{mijt})$ |
| $\log(\text{exports}_{ijt})$                                           | 0.560***<br>(0.0267)        | 0.610***<br>(0.0367)        | 0.566***<br>(0.0227)               | 0.567***<br>(0.0228)               |
| $\log(\text{distance}_{ij})$                                           | 0.306***<br>(0.0973)        | 0.429***<br>(0.0974)        | 0.346***<br>(0.0789)               | 0.345***<br>(0.0789)               |
| rule of law <sub>it</sub>                                              | 7.299***<br>(2.142)         | 8.439***<br>(3.172)         | 6.301***<br>(1.902)                | 7.259***<br>(2.134)                |
| rule of law <sup>2</sup> <sub>it</sub>                                 | −6.592***<br>(1.593)        | −4.917**<br>(2.310)         | −6.022***<br>(1.277)               | −6.778***<br>(1.448)               |
| DC dummy <sub>mijt</sub>                                               |                             |                             | −2.776***<br>(0.361)               | −1.985*<br>(1.146)                 |
| rule of law <sub>it</sub> * DC dummy type <sub>mijt</sub>              |                             |                             | 4.027***<br>(0.530)                | 1.389<br>(3.536)                   |
| rule of law <sup>2</sup> <sub>it</sub> * DC dummy type <sub>mijt</sub> |                             |                             |                                    | 1.981<br>(2.563)                   |
| $\log(\text{GDP per cap}_{it})$                                        | −1.068*<br>(0.576)          | −0.531<br>(0.665)           | −0.956**<br>(0.424)                | −0.963**<br>(0.424)                |
| $\log(\text{GDP per cap}_{it})^2$                                      | 0.0544<br>(0.0357)          | 0.0350<br>(0.0419)          | 0.0526**<br>(0.0257)               | 0.0532**<br>(0.0257)               |
| fin. development <sub>it</sub>                                         | 0.624***<br>(0.134)         | 0.383***<br>(0.124)         | 0.525***<br>(0.104)                | 0.525***<br>(0.104)                |
| LC requirement <sub>it</sub>                                           | 0.553***<br>(0.209)         | 0.296<br>(0.193)            | 0.456***<br>(0.171)                | 0.456***<br>(0.171)                |
| contiguity <sub>ij</sub>                                               | 0.0984<br>(0.108)           | 0.571***<br>(0.176)         | 0.303***<br>(0.103)                | 0.300***<br>(0.103)                |
| common colonizer <sub>ij</sub>                                         | 0.164<br>(0.148)            | −0.00874<br>(0.174)         | 0.0954<br>(0.126)                  | 0.0933<br>(0.125)                  |
| common language <sub>ij</sub>                                          | 0.479***<br>(0.119)         | 0.690***<br>(0.115)         | 0.586***<br>(0.0888)               | 0.584***<br>(0.0889)               |
| common legal origin <sub>ij</sub>                                      | 0.0905<br>(0.0625)          | 0.238***<br>(0.0788)        | 0.160***<br>(0.0487)               | 0.159***<br>(0.0488)               |
| time difference <sub>ij</sub>                                          | −0.0604***<br>(0.0202)      | −0.0653***<br>(0.0226)      | −0.0609***<br>(0.0144)             | −0.0609***<br>(0.0144)             |
| Observations                                                           | 49,404                      | 35,437                      | 84,841                             | 84,841                             |
| R-squared                                                              | 0.625                       | 0.598                       | 0.592                              | 0.592                              |

Note: This table analyzes the relationship between the use of letters of credit and documentary collections and destination country rule of law at the intensive margin.  $\log(\# \text{ LC}_{ijt})$  and  $\log(\# \text{ DC}_{ijt})$  are the logs of the number of MT700 and MT400 messages sent from country  $j$  to country  $i$ , respectively, in year  $t$ .  $\log(\# \text{ SWIFT}_{mijt})$  is the log of the number of SWIFT messages of type  $m \in \{\text{LC}, \text{DC}\}$ . DC dummy<sub>mijt</sub> takes a value of 1 if the message refers to a DC and a value of 0 if the message is related to an LC. All regressions include exporter  $\times$  year fixed effects. Standard errors are in parentheses and are clustered by importing country. \*, \*\* and \*\*\* denote significance at the 10%, 5% and 1% level.

### 5.1.3. The share of LCs in total SWIFT traffic

To test more directly whether LCs are used more for exports to riskier destinations than DCs, we compute the share of LC messages in total SWIFT messages (LC+DC) to destination  $c$  in year  $t$  and regress this variable on the importer's rule of law index and other controls. Regression results are presented in Table 6. Column (1) includes exporter-year-fixed effects only. In column (2), the importer's rule of law is added as an explanatory variable. Its addition increases the  $R^2$  of the regression by 11 percentage points. At the same time, the associated coefficient is highly significant and negative, indicating that the share of LCs in total SWIFT traffic decreases as the ability to enforce a contract in the importing country increases. Column (3) shows that this result is robust to the inclusion of a battery of country variables, which together add only little explanatory power.<sup>25</sup>

Columns (4) to (6) repeat the exercise for source country risk. A stronger rule of law in the exporting country also leads to a smaller fraction of LC messages in total trade finance messages. Note, however, that the effect of higher source country risk is considerably smaller than that of destination country risk. An increase in the

importer's rule of law index of 0.1 decreases the share of LCs in total traffic by around 30 percentage points. In contrast, the same increase in the exporter's rule of law lowers the share by only 15 percentage points. Furthermore, the inclusion of source country risk only increases the  $R^2$  of the regression by 0.7 percentage point and so explains a much smaller fraction of the variation in LC shares across countries than destination country risk.

### 5.1.4. Extensive margin

Source and destination country rule of law also explain the extensive margin of firms' use of LCs and DCs in international trade. Table 7 presents the results from logit regressions, in which the dependent variable takes the value 1 if there was LC or DC traffic between a country pair in year  $t$  and zero otherwise.<sup>26</sup> Controls and clustering are as in the baseline regressions. The probability that a country pair uses LCs and DCs is also hump-shaped in the exporter's and importer's rule of law and increases in the financial development of both countries.<sup>27</sup>

<sup>25</sup> Variables included in the regression but not displayed in columns (3) and (6) of Table 6 are: GDP per capita, GDP per capita squared, financial development, a dummy for a documentation requirement, the number of time zones between country pairs, dummies for contiguity, a common official language and a common colonizer. Note also that when rule of law squared is added as a regressor, its coefficient is insignificant, which suggests that the share of LC messages indeed decreases linearly with the rule of law index.

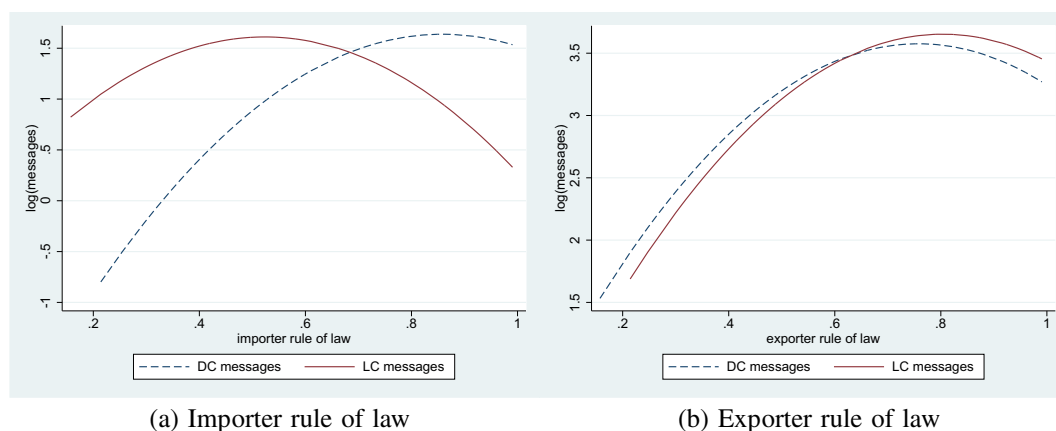
<sup>26</sup> Note that country pairs are defined unidirectionally, so U.S. exports to China are counted separately from Chinese exports to the United States.

<sup>27</sup> This makes sense. LCs are not very attractive to use for trade with the low-risk and high-risk countries, leading to a low level of demand, as we discuss in more detail in the theory section. As establishing and maintaining a network to process LCs is costly for banks, low demand may give rise to a situation where, for some country pairs, banks do not offer LCs at all.

**Table 5**  
LCs, DCs and source country rule of law: intensive margin.

|                                                           | (1)                         | (2)                         | (3)                                | (4)                                |
|-----------------------------------------------------------|-----------------------------|-----------------------------|------------------------------------|------------------------------------|
|                                                           | $\log(\# \text{ LC}_{ijt})$ | $\log(\# \text{ DC}_{ijt})$ | $\log(\# \text{ messages}_{mijt})$ | $\log(\# \text{ messages}_{mijt})$ |
| $\log(\text{exports}_{ijt})$                              | 0.711***<br>(0.0329)        | 0.633***<br>(0.0352)        | 0.643***<br>(0.0308)               | 0.643***<br>(0.0309)               |
| $\log(\text{distance}_{ij})$                              | 0.385***<br>(0.0847)        | 0.417***<br>(0.105)         | 0.358***<br>(0.0835)               | 0.358***<br>(0.0835)               |
| rule of law <sub>jt</sub>                                 | 9.395***<br>(2.462)         | 9.114***<br>(2.833)         | 9.184***<br>(2.334)                | 10.47***<br>(2.577)                |
| rule of law <sup>2</sup> <sub>jt</sub>                    | −6.336***<br>(1.847)        | −5.232**<br>(2.060)         | −5.718***<br>(1.718)               | −6.690***<br>(1.946)               |
| DC dummy <sub>mijt</sub>                                  |                             |                             | 0.284<br>(0.240)                   | 1.223*<br>(0.653)                  |
| rule of law <sub>jt</sub> * DC dummy type <sub>mijt</sub> |                             |                             | −0.477<br>(0.328)                  | −3.600*<br>(2.072)                 |
| rule of law <sup>2</sup> <sub>jt</sub> * DC dummy type    |                             |                             |                                    | 2.336<br>(1.534)                   |
| Observations                                              | 50,181                      | 35,615                      | 85,796                             | 85,796                             |
| R-squared                                                 | 0.624                       | 0.592                       | 0.539                              | 0.540                              |

Note: This table analyzes the relationship between the use of letters of credit and documentary collections and source country rule of law at the intensive margin.  $\log(\# \text{ LC}_{ijt})$  and  $\log(\# \text{ DC}_{ijt})$  are the logs of the number of MT700 and MT400 messages sent from country  $j$  to country  $i$ , respectively, in year  $t$ .  $\log(\# \text{ SWIFT}_{mijt})$  is the log of the number of SWIFT messages of type  $m \in \{\text{LC}, \text{DC}\}$ . DC dummy<sub>mijt</sub> takes a value of 1 if the message refers to a DC and a value of 0 if the message is related to an LC. All regressions control for the log of GDP per capita, the log of GDP per capita squared, LC requirements and the log of financial development in the source country, as well as bilateral controls. All regressions include importer  $\times$  year fixed effects. Standard errors are in parentheses and are clustered by exporting country in columns. \*, \*\* and \*\*\* denote significance at the 10%, 5% and 1% levels.



**Fig. 7.** Estimated relationship between the use of LCs and DCs and the rule of law. Note: The figure illustrates how the use of letters of credit and documentary collections varies with the rule of law of the importer on the left and of the exporter on the right, based on the estimated coefficients displayed in column (3) of Tables 4 and 5, respectively. The solid lines show the log number of LC messages. The dashed lines depict the log number of DC messages.

## 5.2. Additional results and robustness

### 5.2.1. Trade in services

LCs and DCs are instruments that were originally created to facilitate the shipment of physical goods. Accordingly, these instruments are largely used for international trade in goods. However, LCs and DCs also seem to be used occasionally for trade in services. When we control for bilateral trade in services from the World Banks' Trade in Services Database, the associated coefficients are positive and highly significant for both LC and DC messages, indicating that LCs and DCs are employed for services trade, although to a much smaller extent than for trade in goods (see online appendix Table B.1).

### 5.2.2. Related party trade

A significant fraction of international trade is intra-firm, that is, between two affiliated companies (see Bernard et al. 2009). Since the incentives between related parties are typically more aligned, an intra-firm transaction should be less risky than an inter-firm transaction. Accordingly, firms should rely less on banks' trade finance

services in the former case. Information on the share of intra-firm trade is not available worldwide, but the United States Census Bureau provides such data for U.S. transactions. Including the share of U.S. intra-firm exports in our baseline regressions, we find that U.S. exports to destinations that receive a higher share of intra-firm exports employ considerably fewer LCs (see online appendix Table B.2). The effect of intra-firm trade on DCs is not significant.<sup>28</sup>

### 5.2.3. More robustness

We conduct additional exercises to show that our results are robust. We control for the cost of importing and exporting as well as financial development squared. We also drop China, India and South Korea from the sample, countries that exhibit high LC shares for historical reasons. Results also hold when regressions are run on cross-sections of different years. The hump-shaped relationship between the use of LCs and destination country risk even survives

<sup>28</sup> We also ran regressions on U.S. imports including the share of intra-firm imports as explanatory variables. The associated coefficient was not significant.

**Table 6**  
Share of letters of credit in bank-intermediated trade.

|                              | (1)    | (2)                   | (3)                    | (4)    | (5)                    | (6)                     |
|------------------------------|--------|-----------------------|------------------------|--------|------------------------|-------------------------|
| LC share <sub>ijt</sub>      |        |                       |                        |        |                        |                         |
| log(exports <sub>ijt</sub> ) |        |                       | −0.00689*<br>(0.00361) |        |                        | 0.00530***<br>(0.00192) |
| log(distance <sub>ij</sub> ) |        |                       | −0.0228**<br>(0.0103)  |        |                        | −0.0211***<br>(0.00615) |
| rule of law <sub>it</sub>    |        | −0.357***<br>(0.0356) | −0.308***<br>(0.0966)  |        |                        |                         |
| rule of law <sub>jt</sub>    |        |                       |                        |        | −0.0843***<br>(0.0231) | −0.128**<br>(0.0503)    |
| Exporter × Year FE           | Yes    | Yes                   | Yes                    | No     | No                     | No                      |
| Importer × Year FE           | No     | No                    | No                     | Yes    | Yes                    | Yes                     |
| Observations                 | 30,439 | 30,439                | 30,439                 | 30,659 | 30,659                 | 30,659                  |
| R-squared                    | 0.151  | 0.264                 | 0.277                  | 0.315  | 0.322                  | 0.335                   |

Note: This table shows how the importance of letters of credit relative to documentary collections changes with rule of law. The dependent variable LC share<sub>ijt</sub> is the share of letter of credit messages (MT700) in total trade finance messages sent (MT700+MT400) in year *t*. Columns (1) to (3) explore the role of destination country characteristics. Columns (4) to (6) study the role of source country characteristics. Columns (3) and (6) control for the log of GDP per capita, the log of GDP per capita squared, LC requirements and the log of financial development in the destination and source country, respectively as well as bilateral controls. None of the coefficients of these controls are significantly different from zero. Standard errors are in parentheses. They are clustered by importing country in columns (1) to (3) and by exporting country in columns (4) to (6). \*, \*\* and \*\*\* denote significance at the 10%, 5% and 1% levels.

when different trade finance data from U.S. banks' regulatory filings available at the Federal Reserve are used. All regression tables for these exercises can be found in online Appendix B.

We also address the worry that exports could be correlated with unobserved factors that also affect firms' choices of payment contracts, generating an endogeneity problem. We would ideally want to run regressions on the share of LC trade over total trade. But this is not feasible for the bilateral data set with many exporting and importing countries because we observe message counts and not values for most of the sample period. At the same time, we only know export values and not the number of export transactions. Fortunately, more detailed information is available for the United States. We normalize the left-hand-side variable by dividing the number of LC (DC) messages by the number of shipments from the United States to the respective importing country; the new dependent variables reflect the shares of U.S. shipments that use LCs and DCs. These are regressed on the various explanatory variables as before (see columns (3) and (4) of Table B.2).

In an additional exercise, we compute the share of U.S. exports that use LCs and DCs based on the value data from SWIFT for 2011 and 2012 and estimate equation (1) (see columns (5) to (6) of Table B.2). While the normalizations naturally change the magnitudes of the coefficients, and standard errors become larger when regressions are based on only two cross-sections resulting in much fewer observations, the results imply the same qualitative relationship as identified in the baseline regressions, although the rule-of-law coefficients are not significant at standard significance levels for the regression of DC shares.

## 6. A model of payment choice

In this section, we lay out a model that rationalizes the documented empirical patterns. We derive results on transaction sizes, the role of source and destination country risk and the extensive margin. We also calibrate and simulate the model to show that it can account for the hump-shaped relationship between the use of LCs (DCs) and destination country risk. The model can also generate the observed increase in LC transactions after the Lehman collapse.

### 6.1. The different payment contracts

In the model, one exporter is matched with one importer. They play a one-shot game and have a choice between four payment

contracts: cash-in-advance (CIA), open account (OA), documentary collection (DC) and letter of credit (LC). Both firms are risk-neutral. The exporter has all bargaining power and makes a take-it-or-leave-it offer to the importer specifying the payment contract, the price and the quantity of the goods to sell. *R* denotes the sales value of the goods in the destination country and *K* the production costs in the source country.

Firms are either good or bad. A good firm always fulfills a contract. A bad firm breaks it whenever it is profitable. The share of good firms in the source country is given by  $\eta$  and in the destination country by  $\eta^*$ . If a firm does not voluntarily fulfill a contract, its trading partner can attempt to enforce it in court. This is successful in the source country with probability  $\lambda$  and in the destination country with probability  $\lambda^*$ .

#### 6.1.1. Cash-in-advance

Under cash-in-advance terms, the importer first pays for the goods and then the exporter delivers them. The exporter can decide to keep the money without delivering the goods. Exporters that are of the good type (share  $\eta$ ) always fulfill the contract, whereas bad exporters (share  $1 - \eta$ ) always try to get away without producing. If an exporter defaults on the contract, the importer goes to court, which enforces the contract with probability  $\lambda$ . The exporter sets the price of the goods *C* to maximize its expected profit.<sup>29</sup> It also respects the participation constraint of the importer. The expected profit of a good exporter can be derived as:

$$E[\Pi_E^{CIA}] = (\eta + (1 - \eta)\lambda)R - K. \quad (3)$$

The profitability of cash-in-advance terms increases in the degree of contract enforcement  $\lambda$  and the share of good firms  $\eta$  in the source country.

#### 6.1.2. Open account

With an open account, the exporter first sends the goods and then the importer pays for them. Good importers (share  $\eta^*$ ) pay

<sup>29</sup> Two bad-exporter strategies need to be distinguished: pooling and separating. Under pooling, a bad exporter imitates the good exporter. Under separating, it chooses a different strategy that reveals its type. Following Schmidt-Eisenlohr (2013), we assume that conditions are such that only the pooling case arises. This is the case if  $\frac{R}{K} > \frac{1}{\eta}$ . See Appendix D for details and a formal derivation of all results.

**Table 7**  
LCs, DCs and rule of law: extensive margin.

|                                        | (1)                  | (2)                  | (3)                  | (4)                  |
|----------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                        | $D_{ijt}^{LC}$       | $D_{ijt}^{DC}$       | $D_{ijt}^{LC}$       | $D_{ijt}^{DC}$       |
| log(exports <sub>ijt</sub> )           | 0.635***<br>(0.0282) | 0.650***<br>(0.0339) | 0.733***<br>(0.0286) | 0.665***<br>(0.0306) |
| log(distance <sub>ij</sub> )           | −0.174<br>(0.123)    | 0.585***<br>(0.131)  | −0.123<br>(0.118)    | 0.401***<br>(0.138)  |
| rule of law <sub>it</sub>              | 7.484***<br>(2.080)  | 12.65***<br>(3.446)  |                      |                      |
| rule of law <sup>2</sup> <sub>it</sub> | −6.068***<br>(1.630) | −6.640**<br>(2.760)  |                      |                      |
| fin. development <sub>it</sub>         | 0.896***<br>(0.150)  | 0.389**<br>(0.160)   |                      |                      |
| rule of law <sub>jt</sub>              |                      |                      | 8.650***<br>(2.160)  | 9.482***<br>(2.843)  |
| rule of law <sup>2</sup> <sub>jt</sub> |                      |                      | −5.405***<br>(1.744) | −3.985*<br>(2.254)   |
| fin. development <sub>jt</sub>         |                      |                      | 0.808***<br>(0.149)  | 0.549***<br>(0.157)  |
| Exporter × Year FE                     | Yes                  | Yes                  | No                   | No                   |
| Importer × Year FE                     | No                   | No                   | Yes                  | Yes                  |
| Observations                           | 100,144              | 100,958              | 101,266              | 92,070               |
| Pseudo R-squared                       | 0.528                | 0.529                | 0.538                | 0.537                |

Note: This table analyzes the relationship between the use of letters of credit and documentary collections and rule of law at the extensive margin.  $D_{ijt}^{LC}$  and  $D_{ijt}^{DC}$  are dummy variables that take the value of 1 if the importer-exporter pair  $ij$  had positive LC and DC traffic, respectively, in year  $t$ . Columns (1) and (2) explore the role of destination country characteristics. Columns (3) and (4) study the role of source country characteristics. Columns (1) and (2) (3 and 4) control for the log of GDP per capita, the log of GDP per capita squared, LC requirements and the log of financial development in the destination (source) country. Standard errors are in parentheses. They are clustered by importing country in columns (1) and (2) and by exporting country in columns (3) to (4). \*, \*\* and \*\*\* denote significance at the 10%, 5% and 1% levels.

the agreed price in any case, while bad importers (share  $1 - \eta^*$ ) try to get away without paying. When faced with a bad importer, the exporter goes to court, which enforces the contract with probability  $\lambda^*$ . The exporter maximizes expected profit respecting the participation constraint of the importer, which results in:

$$E[\Pi_E^{OA}] = (\eta^* + (1 - \eta^*)\lambda^*)R - K. \quad (4)$$

Because the commitment problem is on the importer's side, better contract enforcement and a higher share of good firms in the destination country (higher  $\lambda^*$  and  $\eta^*$ ) increase the profitability of an open account.

### 6.1.3. Documentary collection

In a documentary collection transaction (Fig. 2), banks handle documents that transfer the ownership rights from the seller to the buyer. A DC ensures that the importer receives the documents only after paying for the goods. Because the importer typically needs the documents to fully employ the delivered goods, a DC improves the reliability of payment compared with an open account.<sup>30</sup> We make two assumptions to capture these features of a documentary collection. First, we assume that with a documentary collection fewer firms try to cheat. More specifically, the share of firms that voluntarily fulfill their contracts increases by a factor of  $1 + \phi^{DC} \in (1, \frac{1}{\eta^*})$ . Payment then happens with probability  $\eta^*(1 + \phi^{DC}) + (1 - \eta^*(1 + \phi^{DC}))\lambda^*$ . More generally, define the probability of payment as a function of  $\phi$  as:

$$\Lambda(\phi) = \eta^*(1 + \phi) + (1 - \eta^*(1 + \phi))\lambda^*. \quad (5)$$

<sup>30</sup> Even with a DC arrangement, however, the importer may still not pay. For example, when the importer can take possession of the goods and divert them without demonstrating legal ownership. Alternatively, the importer may not pay because it no longer wants the goods or because it simply does not have the funds (for example in bankruptcy). We abstract from the latter case in the analysis here.

Second, the exporter needs to pay a fixed transaction fee  $F^{DC}$  to the bank for its handling of the documentary collection. Expected profits are then as follows:

$$E[\Pi_E^{DC}] = \Lambda(\phi^{DC})R - F^{DC} - K. \quad (6)$$

Relative to an open account, the exporter's expected profit increases due to the higher probability of being paid. However, the profit is reduced by the fee  $F^{DC}$ , which the exporter pays at the beginning of the transaction. Note that the fee is not proportional to the value of the trade so that a documentary collection features increasing returns to scale.

### 6.1.4. Letter of credit

When a trade is settled with an LC (Fig. 2), banks do not only hand over documents to the importer as in a DC but they may also advance the importer's payment. Banks therefore have a strong incentive to screen and monitor importers. Accordingly we assume that the share of importers that try to get away without paying decreases by more with an LC than with a DC. Specifically, we assume that the share of importers that always pay increases by a factor of  $\phi^{LC} > \phi^{DC}$ . There remains a risk that the bank does not get paid, which happens with probability  $1 - \Lambda(\phi^{LC})$ . At the same time, the fixed fee that the bank charges for an LC to cover screening, monitoring and document handling costs is higher than for a DC. It is given by  $m > F^{DC}$  and is independent of the value of the transaction. Under perfect competition, banks charge an importer the monitoring cost  $m$  plus the expected loss from extending the guarantee. The letter of credit fee is thus given by:

$$F^{LC} = m + (1 - \Lambda(\phi^{LC}))C^{LC}. \quad (7)$$



The exporter's expected profits with an LC can be derived as:

$$\Pi^{LC} = \frac{1 - (m/R)}{[2 - \Lambda(\phi^{LC})]} R - K. \quad (8)$$

Because the risk that the exporter does not deliver is eliminated by an LC, expected profits are independent of the enforcement probability  $\lambda$ . However, profits depend on the enforcement probability  $\lambda^*$  in the importing country through the fee  $F^{LC}$ . Recall that the probability  $\Lambda(\phi)$  that the importer pays increases in  $\lambda^*$ . Thus, under this payment form, the higher the risk that the importer does not pay, the lower the profits are. As with a DC, the LC case contains an element of increasing returns to scale since the monitoring cost  $m$  is fixed. The average cost of an LC decreases in transaction size  $R$ .

## 6.2. Results

### 6.2.1. Transaction sizes

As discussed above, both DCs and LCs imply fixed document handling, screening and monitoring costs that give rise to increasing returns to scale. Therefore, the higher the value of a contract is, the more attractive DCs and LCs become. Fixing the ratio  $R/K$  as well as enforcement probabilities  $\lambda$  and  $\lambda^*$ , it is easy to show that for small transaction sizes (low  $R$ ), firms should rely on OA or CIA to save on fixed costs. Transactions with intermediate values should be settled with DCs. When transactions are very large, LCs are most attractive; they imply the highest fixed costs but at the same time reduce payment risk the most.<sup>31</sup> These results perfectly match the empirical patterns uncovered in the data.

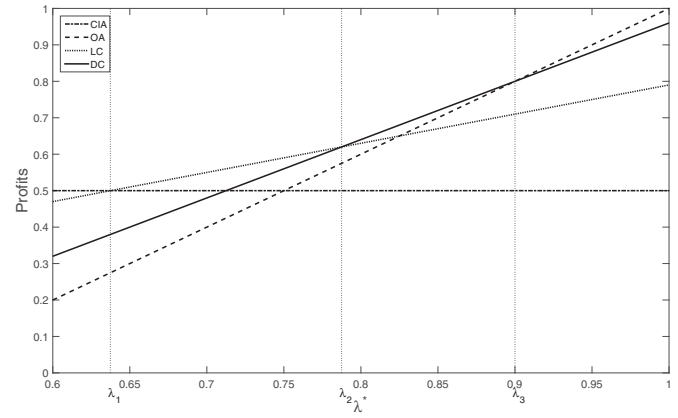
### 6.2.2. Importer risk

The payment forms differ in how risk is allocated between trading partners (see again Table 1 for a summary). Fig. 8 illustrates the relationship between the optimal payment contract and destination country risk. It plots the exporter's expected profits under OA, CIA, LC and DC as a function of  $\lambda^*$  keeping all other parameters constant (including the transaction size). Recall that under each payment form the exporter is differently exposed to the risk that the importer does not pay. The higher the exporter's exposure, the larger is the effect of a change in the probability  $\lambda^*$  on expected profits, and the steeper is the profit line in the figure.

OA allocates all the risk to the exporter. Profits under this payment form therefore respond the most to changes in  $\lambda^*$ . Since a DC and an LC reduce the risk that the importer gets away without paying, expected profits with an LC or a DC are less responsive to changes in  $\lambda^*$ . When the trade is settled on CIA terms, the commitment problem on the importer's side is eliminated. Therefore, CIA profits are independent of the degree of contract enforcement in the destination country, resulting in a flat line in the figure.

The ordering of the slopes is unique and implies that if each payment contract is optimal for some value of  $\lambda^*$ , then CIA is chosen for destination countries with the weakest contract enforcement.<sup>32</sup> As the probability that a contract is enforced rises, first LCs dominate, then DCs. An open account is optimal when the risk that the importer does not pay is particularly low. In the figure, the optimal payment form switches at  $\lambda_1$ ,  $\lambda_2$  and  $\lambda_3$ .

As we show below, the unique ordering leads to a hump-shaped relationship between the use of LCs and DCs and destination country risk when there is a random shock to the profitability of the different payment forms that varies across trading partners.



**Fig. 8.** Profits as a function of contract enforcement in the destination country. Note: The graph plots the expected exporter profits under the four payment forms (cash-in-advance, open account, documentary collection and letter of credit) as a function of contract enforcement  $\lambda^*$  in the destination country. Parameters are:  $\lambda^* \in [0.6, 1]$ ,  $\lambda = 0.5$ ,  $R = 2$ ,  $K = 1$ ,  $F^{LC} = 0.21$ ,  $F^{DC} = 0.04$ ,  $\phi^{LC} = 0.6$ ,  $\phi^{DC} = 0.2$ .

### 6.2.3. Exporter risk

The empirical analysis showed that the use of LCs and DCs is not only hump-shaped in destination but also in source country contract enforcement. It is possible to extend the model to generate this result. For an illustration, consider the choice between CIA, OA and LC. A hump-shaped relationship between the use of LCs and source country contract enforcement arises if the profitability of LCs decreases with source country risk.

Why might this be the case? One possible explanation is related to the document handling of LCs. Sometimes, LCs get rejected by the issuing bank for formal reasons, for example, because paperwork is incomplete or contains errors. This problem is likely to be bigger when legal institutions in the source country are weak, making LCs more costly and risky in this case.<sup>33</sup> Another possible explanation is that source country contract enforcement matters because it correlates with aspects of the country's banking system like the risk that banks fail or the probability that banks do not fulfill their obligations to foreign counter-parties. If low rule of law countries had less reliable banking sectors, LC and DC profits would positively correlate with source country enforcement.

To incorporate these aspects into the model, assume that LC profits increase in the source country enforcement probability  $\lambda$ :

$$E[\Pi_E^{LC}] = (\lambda^* + (1 - \lambda^*)\phi^{LC})R - K - F^{LC} - \delta R(1 - \lambda), \quad (9)$$

where  $\delta < 1$ . Then, if all payment forms are used for some level of  $\lambda$ , the use of LCs first increases and then decreases in  $\lambda$ . The intuition is similar to before. Profits under OA are independent of  $\lambda$ , while CIA profits decrease in this variable. The derivative of LC profits with respect to  $\lambda$  is smaller than the derivative of CIA profits with respect to  $\lambda$ . As a result, LCs are employed the most for intermediate levels of  $\lambda$ . When source country risk is low, firms can avoid the fees associated with LCs by choosing CIA. When source country risk is high, they settle transactions with OA.

## 6.3. Calibration and simulations

In this section, we calibrate the model to moments on the use of LCs and DCs in U.S. exports between 2011 and 2013. We show

<sup>31</sup> Results on transaction sizes are proven formally in Appendix D. See Proposition 2.

<sup>32</sup> This is proven formally in Appendix D. See Proposition 1.

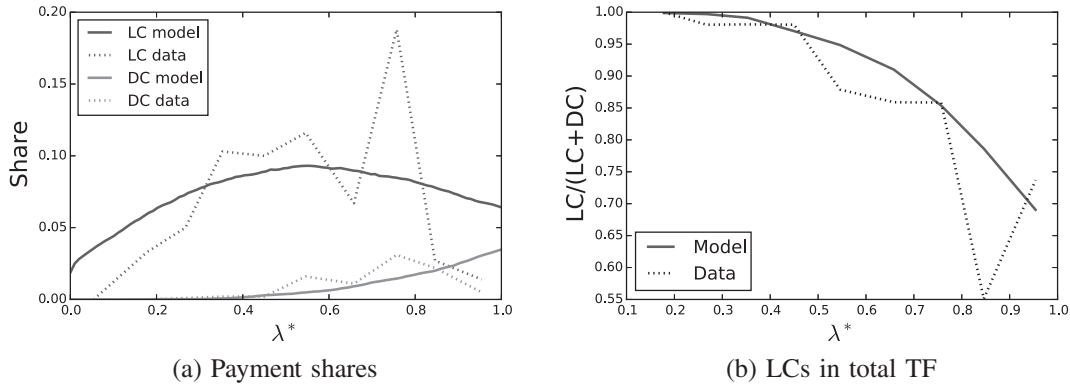
<sup>33</sup> The same argument applies to DCs. This might explain why the relationship to exporter risk is very similar for DCs and LCs.

**Table 8**

Data moments targeted in the calibration.

| Rule of law bucket $i$ | (1) | (2) | (3) | (4)  | (5)  | (6)  | (7) | (8)  | (9)  | (10) |
|------------------------|-----|-----|-----|------|------|------|-----|------|------|------|
| LC share (%)           | 0   | 3.3 | 5.0 | 10.3 | 10.0 | 11.6 | 6.7 | 18.8 | 2.7  | 1.4  |
| DC share (%)           | 0   | 0   | 0.1 | 0.2  | 0.2  | 1.6  | 1.1 | 3.1  | 2.2  | 0.5  |
| Share of exports (%)   | 0   | .5  | 1.3 | 3.1  | 32.1 | 8.8  | 4.9 | 5.3  | 11.4 | 32.6 |

Note: This table summarizes the information used in the calibration exercise. For the exercise, all destination countries are grouped into 10 bins according to their rule of law. The table reports the average rule of law of countries in each bin, the share of U.S. exports going to those countries as well as the share of U.S. exports to those countries that use LCs and DCs, respectively.



**Fig. 9.** Model predictions and data. Note: The charts plot moments derived from the calibrated model (solid lines) as well as from the data (dashed lines). Panel (a) shows for each range of  $\lambda^*$  the actual (data) and predicted (model) shares of LCs and DCs in U.S. exports to countries that fall into that range of destination country contract enforcement. Panel (b) plots the actual and predicted shares of LCs in total trade finance (LCs + DCs) for the same rule-of-law intervals.

that the model can replicate key features of the data quantitatively, including trade finance choices during the financial crisis. While we acknowledge that there are multiple ways to bring the model to the data, we present one successful way here.

For the calibration, we slightly modify the model. Specifically, we introduce a multiplicative error term to the profitability of each payment contract. For example, profits from cash-in-advance become:

$$\left( \frac{\eta + (1 - \eta)\lambda}{1 + r^*} R - K \right) * \epsilon, \quad (10)$$

where  $\epsilon$  is drawn i.i.d. from a normal distribution with mean 1 and variance 0.1.<sup>34</sup>

We calibrate the model to match 22 moments regarding the use of LCs and DCs. For each payment form (LC and DC), we calculate its share in aggregate U.S. exports (2 moments). In addition, we create ten evenly-spaced destination country rule of law buckets and calculate the shares of LCs and DCs within each bucket (20 moments).<sup>35</sup> Table 8 reports these shares for the ten buckets.

For the calibration, we need values for  $\lambda$  and  $\lambda^*$ . Instead of taking the normalized rule of law variable at face value, we use the following mapping:  $\lambda^* = (\bar{\lambda} * \text{law})^{0.5}$ , with a baseline value of  $\bar{\lambda} = 1$ . This mapping is not necessary but improves the fit of the model and we can later vary the parameter  $\bar{\lambda}^*$  to simulate a crisis episode.

<sup>34</sup> The multiplicative error terms are necessary to create some variation in the choice of payment forms that is driven by factors outside of the model. This allows to match the smooth distributions in the data, where for most rule of law values each payment form is used by at least some firms.

<sup>35</sup> That is, for bucket 1, take all countries with rule of law between 0 and 0.1 and calculate the ratio of the total value of all LCs (DCs) over total exports.

Setting a few parameters exogenously ( $\eta = \eta^* = 0.7$ ,  $\phi^{DC} = 0.1$ ,  $R = 2$  and  $K = 1$ ), we select the fixed costs of a DC,  $F^{DC}$ , the letter of credit monitoring cost,  $m$ , and the LC parameter,  $\phi^{LC}$ , to minimize:<sup>36</sup>

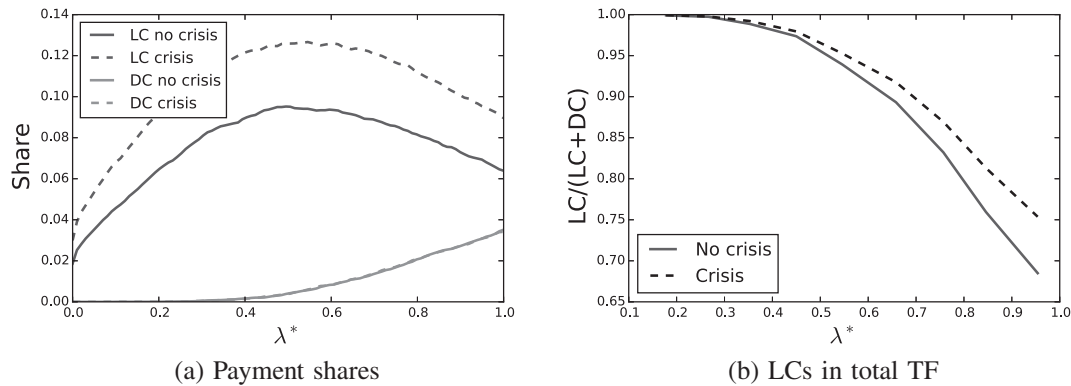
$$\begin{aligned} \min_{F^{DC}, m, \phi^{LC}} \text{Error} = & \omega_{LC} \text{error}_{LC} + \omega_{DC} \text{error}_{DC} + \sum_{i=1}^{10} \omega_{LC,i}^D \text{error}_{LC,i}^D \\ & + \sum_{i=1}^{10} \omega_{DC,i}^D \text{error}_{DC,i}^D, \end{aligned} \quad (11)$$

where  $\text{error}_{LC}$  and  $\text{error}_{DC}$  denote the errors belonging to the shares of LCs and DCs in aggregate U.S. exports.  $\text{error}_{LC,i}^D$  and  $\text{error}_{DC,i}^D$  denote the errors associated with the different rule of law buckets. Each error is calculated as the squared difference between the respective data moment and model moment. Finally, we use the following error weights:  $\omega_{LC} = \omega_{DC} = 1$  and  $\omega_{LC,i}^D = \omega_{DC,i}^D = 2.5$ .

### 6.3.1. Additional calibration details

To compute the 22 model moments, we first need realizations of the multiplicative error to contract profitability  $\epsilon$ . For each contract, we draw  $N = 2,000$  times from the distribution. Discretizing  $\lambda^*$ , we then calculate for each level of  $\lambda^*$ , how often each contract dominates its alternatives and divide by  $N$  to obtain its share in transactions. Finally, we map the  $\lambda^*$  grid back onto the 10 evenly-spaced rule of law buckets and calculate the average shares of LCs and DCs within each bucket. To compute the shares of LCs and DCs in total U.S. exports, we calculate the average shares across all buckets, weighing by the share of exports that goes to destinations in that bucket. As the model does not deliver predictions about aggregate trade flows

<sup>36</sup> We experimented with additionally calibrating  $\eta$ ,  $\eta^*$  or  $\phi^{DC}$ . This slowed down the solution algorithm substantially without noticeably improving the model fit.



**Fig. 10.** Payment choice and financial crisis. Note: The charts plot results from the crisis experiment where  $\bar{\lambda}$  is changed from 1 to 0.94. Solid lines depict the baseline calibration and dashed lines show the outcomes in the crisis scenario. Panel (a) shows for each range of  $\lambda^*$  the baseline and crisis shares of LCs and DCs in U.S. exports to countries that fall into that range of destination country contract enforcement. Panel (b) plots the baseline and crisis shares of LCs in total trade finance (LCs + DCs) for the same rule-of-law intervals.

as a function of destination country rule of law, we take these trade shares from the data (these shares are also reported in Table 8).

### 6.3.2. Baseline calibration

Panel (a) of Fig. 9 shows the payment choices resulting from the calibrated model. The model replicates the hump-shaped relationship between the use of LCs and destination country contract enforcement and also roughly matches the pattern of DCs, although the relationship between DC use and destination country rule of law generated by the model is increasing rather than hump-shaped. Aggregate (trade-weighted) shares are predicted at 8.2% for LCs (data: 6.7%) and 2.3% for DCs (data: 0.8%). Given that we try to match 22 moments with only 3 model parameters, the model fits the data remarkably well. We also compute the share of LCs in total trade finance (LC+DC), a moment not directly targeted in the calibration. As shown in panel (b) of Fig. 9, the model does a good job at explaining the relative importance of letters of credit.

### 6.3.3. Simulating the crisis

Following the Lehman collapse in 2008q3, LC messages over exports increased substantially. For U.S. exports, this ratio rose by 23% from 2008 to 2009. To replicate this change in the model, we introduce an increase in risk in all countries, modeled as a reduction in  $\bar{\lambda}^*$  keeping all other parameter values, including trade shares, constant. We find that  $\bar{\lambda}^*$  needs to drop from 1 to 0.94 to generate the observed increase in LC use. This experiment is illustrated in Fig. 10. While the use of LCs increases a lot following the change in risk, the use of DCs hardly changes (panel(a)).<sup>37</sup> As a consequence and in line with the observed pattern in the data, the share of LCs in overall trade finance (LCs + DCs) increases (panel (b)).

## 7. Summary and implications

Using unique data derived from SWIFT messages, this paper establishes facts on the global use of letters of credit and documentary collections. As of now, it provides the most reliable and comprehensive evidence to researchers and policy makers, who, previously, had to rely on surveys or data from a single firm or a few individual

countries to grasp the relevance of these instruments for international trade. The fact that the SWIFT data line up very well with available customs data underpins their high quality and accuracy.

Several key results emerge from the data. While LCs and DCs are frequently used in world trade, they are used less often than what bank surveys have suggested. LCs are employed the most for intermediate risk countries and not for the riskiest destinations. The 2007/2008 financial crisis led to a sharp increase in the number of LCs sent relative to trade.

We also document that there is a large number of country pairs with zero LC or DC message flows despite positive trade flows. These zeros are consistent with the view of many policy makers that there may be a gap in the provision of trade finance by commercial banks.<sup>38</sup> However, our theory points at an additional reason why LCs may be used little for such destinations. In these countries, the risk of non-payment by buyers tends to be higher, which make LCs more expensive. Firms may therefore decide to use cash-in-advance terms instead of LCs when selling to these destinations.

While the SWIFT data provide insights into the use of LCs and DCs in international trade, there is a lack of similar information for other payment options. Future research will hopefully be able to close this gap by drawing on comprehensive data on the use of cash-in-advance and open-account terms as well as trade credit insurance. To understand how trade finance shapes international trade, a further exploration of how firm and industry characteristics affect firms' international payment choices is also desirable.

## Appendix A. Data appendix

### A.1. Data sources

- SWIFT MT400 and MT700 messages: SWIFT Institute.
- Bilateral trade data: Quarterly trade data is from the IMF's Directions of Trade Statistics; yearly data is obtained by summing over 4 quarters.
- U.S. trade data: Information on the share of related party trade and trade cards (number of shipments) is from the United States Census Bureau.
- Data on services trade: Trade in Services Database from the World Bank.

<sup>37</sup> To match the really low shares of DCs in trade, DCs have to be relatively bad (compared to LCs) in reducing risk. For this reason, their use responds much less to a change in the risk environment. In the simulation, there is an increase in the use of DCs when  $\bar{\lambda}^*$  gets lowered but it is too small to be seen in panel (a) of Fig. 10.

<sup>38</sup> There are ongoing discussions at the WTO about whether the private sector is able to meet the demand for trade finance, especially in poor countries (see Working Group on Trade, Debt and Finance (2014)).

- Rule of law: World Government Indicators provided by the World Bank, normalized as follows:  $law_{new} = \frac{law_{old} - \min(law_{old})}{\max(law_{old}) - \min(law_{old})}$ .
- Nominal GDP per capita: World Development Indicators provided by the World Bank.
- Distance and other gravity variables: CEPII. For a description, see Mayer and Zignago (2005).
- Financial development: proxied by private credit by deposit money banks over GDP from Financial Structure Database provided by the World Bank. For a description, see Beck et al. (2009).
- Letter of credit requirements: Information on whether a country requires that importers use letters of credit is from the IMF's Annual Report on Exchange Arrangements and Exchange Restrictions (ARERER).
- U.S. GDP deflator: FRED Economic Data provided by the Federal Reserve Bank of St. Louis.
- Cost to import and cost to export: Doing Business Indicators provided by the World Bank, USD per container.

#### A.2. Offshore centers and tax havens (excluded)

Antigua and Barbuda, Azerbaijan, Netherlands Antilles, United Arab Emirates, Bahrain, Bahamas, Belize, Bermuda, Barbados, Cayman Islands, Cyprus, Gibraltar, Grenada, Hong Kong, Oman, Ireland, Jordan, Lebanon, Macao, Monaco, Maldives, Malta, Mauritius, Saint Lucia, Seychelles, Singapore, Switzerland, Taiwan, Vanuatu, Samoa.

#### Appendix B. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.jinteco.2017.03.007>.

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